

To understand a concept, a person must understand some collection of information and a framework by which to interpret the information. With a given amount of time and effort to invest, I think of efficiently learning as a 2 step process:

(1) Consider broadly the relative importance of different pieces of information and ordering (does learning certain information *first* make it easier to learn other information) and consider the relative importance of different elements of a framework. As an example, in calculus, understanding the spatial interpretation of a derivative is more important in a high school applied physics setting than understanding hyperbolic trigonometry.

(2) Take steps towards learning the content, weighted by the relative importance. In this case, spend more time earlier understanding derivative interpretation and less time later on hyperbolic trig. The former is a central focal point of understanding the subject as a whole and the latter is a tangential side-topic of little practical relevance.

I think people, or at least students in high school, allocate far too much time and effort in the second step and far too little in the first. There are diminishing returns to the first, but I think many people move on before they've even gotten close to an optimal allocation, and then proceed in a deeply inefficient way.

In many aspects of life a little bit of foresight is worth a lot of effort, and I think learning is one where this should be more emphasized. I feel that even a 10 minute discussion of this at the beginning of certain high school classes could make a world of difference to students that are working hard but still struggling to succeed.

The concept of a “solid of revolution” is fundamentally intuitive, and almost requires no degree of abstraction from a foundationally sound understanding of what an integral is. I have often seen people caught up in learning this concept, and they can’t wrap their head around it. A good approach here would be: go back, and spend more time understanding what an integral is. And then understand what a circle is. Understand how to add and subtract functions. Then it will become clear the way in which the volume of a solid of revolution relates to certain properties of the underlying integral. All of these concepts are important and will provide insight into other topics. As a bonus, you simply get solids of revolution for free.

There’s no need to spend time learning *how* to do solids of revolution, and literally memorizing steps to follow in order to do the process, and in which situations to do which steps. All of this time is wasted because it’s mechanically forcing something that is a direct byproduct of more important concepts. I think of it like watering flowers before it rains. Let it rain first, then you will realize there’s no need to water.

Most topics are not their own thing, they’re just a collection of insights from other more foundational knowledge. This is why ordering is so absolutely crucial, and the core task is to understand the synergistic value of each topic. In this case, how can someone know *before* understanding these topics that they will work together to elucidate solids of revolution?

That sort of insight and thought is precisely the thing that needs to be cultivated - how might this information be useful and apply to other sorts of problems? It is often said that thinking of the right questions is far more important than coming up with the right answers, and I think this is one of those cases.